Employee Churn Prediction: Results Summary

Predictive analytics project on 5,000 e-commerce and retail employees, with a prescriptive extension.

1. Project at a glance

Item	Result	Simple meaning
Goal	Predict who will leave	Find at-risk employees early
Data	5,000 employees, 10 columns	Small dataset, limited information
Leavers	12.3% (613 people)	Very imbalanced: about 1 in 8 leaves
Test group	1,000 people, 123 real leavers	The "exam" the models never saw
Models compared	Logistic Regression, Random Forest, XGBoost, Neural Network	Simple, tree-based and deep learning

2. What each score means

Score	Simple meaning	Good is
Recall	Of the real leavers, how many we caught	High
Precision	Of the people flagged, how many really left	High
F1	One score blending recall and precision	High
ROC-AUC	How often a leaver is ranked riskier than a stayer (0.5 = coin flip)	High
PR-AUC	How well we find leavers (random = 0.12)	High
Brier	How honest the percentages are	Low

3. Model comparison (test set)

Model	ROC-AUC	PR-AUC	Precision	Recall	Plain-English description
Logistic Regression	0.736	0.285	25%	50%	Best at ranking, easy to explain
XGBoost	0.719	0.281	24%	49%	Same performance, harder to explain
Random Forest	0.718	0.275	26%	43%	Most careful, misses the most leavers
Neural Network	0.722	0.270	24%	59%	Catches the most leavers, flags the most people

All four are close. The differences are inside the margin of error, so none is clearly better.

4. Why I recommend each one

Model	Decision	Reason
Logistic Regression	**Use**	Top scores, simple, and you can say exactly why someone is flagged
XGBoost	Backup	Equal performance, useful if you add more data later

Model	Decision	Reason
Random Forest	Skip	Lowest scores, no advantage
Neural Network	Skip	Too little data, and hard to explain

5. What drives people to leave

Factor	Finding	Meaning
Overtime	Over 20 hours a month: 39% leave, under 4 hours: 7%	Heavy overtime is the strongest warning sign
Job satisfaction	Score under 40: 34% leave	Unhappy staff leave more
Job level	Entry level 20%, managers 2%	Junior staff are the most at risk
Department	Fulfillment highest at 19%	Some teams have bigger problems

6. How well the model separates risk (test set)

Risk group	Actual leave rate	Meaning
Top 10% highest risk	31%	Almost 3x the average
Next 20%	21%	Elevated risk
Remaining 70%	7%	Low risk
Top 20% overall	Contains 45% of all leavers	Random picking would find only 20%